Algebra 1 — Arithmetic Sequences & Scatter Plots — Answer Key

Part A — Arithmetic sequences

1. Arithmetic; common difference 5.

2. Not arithmetic — the differences are 5, 4, 3, not constant.

3. Arithmetic; common difference −6.

4. Recursive: $a_1 = 4$, $a_n = a_{n-1} + 7$ · Explicit: $a_n = 4 + 7(n-1)$

5. Recursive: $a_1 = 2.5$, $a_n = a_{n-1} + 0.75$ · Explicit: $a_n = 2.5 + 0.75(n-1)$

6. Recursive: $a_1 = 12$, $a_n = a_{n-1} - 7$ · Explicit: $a_n = 12 - 7(n-1)$

7. Explicit: $a_n = 7 + 4(n-1)$

8. Explicit: $a_n = -4 - 6(n-1)$

9. Recursive: $a_1 = 13$, $a_n = a_{n-1} + 5$

10. Sample answer: use the recursive formula when you only need the next few terms; use the explicit formula when you need a far-out term (like the 100th) directly.

11. Savings: $a_n = 210 + 40n$ (a_1 = \$250 at month 1, then +\$40/month); month 12 = \$690.

Part B — Scatter plots

12. Positive association.

13. Negative association.

14. Positive correlation. Sample trend line: $y = 5x + 6$.

15. Negative correlation. Sample trend line: $y = -6x + 58$.

16. Expected height at 2 weeks ≈ 16 cm — the plant grows about 5 cm per week.

17. Expected sales at \$6.50 ≈ 19 thousand units — each \$1 increase is associated with about 6,000 fewer units sold.

18. No — a plant does not grow at the same rate forever; growth slows and stops as it matures (any sensible real-world limit).